Lesson 2

Recap

Variables: containers for storing values
-Need to follow some rules to name variables

Data Types: different types of data values
int: integer values only (no decimal).
float: decimal values
string: sequence of characters in quotes
boolean : True/False
None : no value assigned

Review

define 3 variables and print them

name, grade, school

User Input

User input is used to:

- make a program more interactive
- gathering data for calculations

We use the input function to do so:

Ex.
name = input(“What is your name: ”)
print(name)

What if we have to take an input that is a number?

Use int or float function to convert the value:

num = int(input(“Enter your number”))
print(num)

grade = float(input(“What is your math grade”))
print(grade)

Operators

-special symbols and keywords used to perform operations on variables and values

ARITHMETIC OPERATORS:
-used to perform standard mathematical operations

Addition(+) : Can be used to add integers, floats and strings
→ explain in vs code

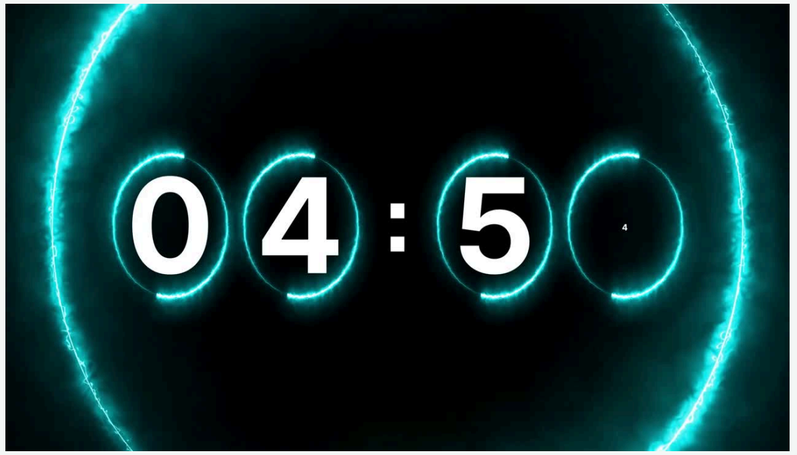
Subtraction(-) : Can be used to subtract integers and floats

Multiplication(*) : Can be used to multiply integers, floats and strings(repeats the string that many times)
→ explain in vs code

Division(/): can be used to divide integers and floats

Write a program that takes an input from the user of 2 numbers and returns the sum of those.

We will discuss after 5 mins



Steps if you get lost:

1) get input of 1st number from user

2) get input of 2nd number from user

3) print the addition of the 2 numbers

Comments

-lines of text used to explain code

Single Line Comments:

Start with # symbol, python interpreter doesn't run anything written as code after this.

Multi Line Comments:

-can use multiple lines to explain code

1) place a # at the start of each line to let the interpreter know this is not code

2) use triple quotes to enclose the entire comment

```
# This is a single-line comment
print("Hello, World!") # This is an inline comment
```

```
# You can write a long explanation
# by using multiple hashtag lines
# one after another.

"""
Or, you can use triple quotes.
Since this string isn't assigned to a variable,
it functions as a multi-line comment.
```



Add comments to the user input program we just wrote.

More Arithmetic Operators

Modulus(%): returns the remainder of division

ex. 15//4 = 3

Exponentiation(**): exponentiates

ex. 5**2 = 25 → 5 to the second power or 5 squared

Floor Division(//): standard division and floor division produce the same whole numbers, but floor division truncates the decimals.

Standard: 13 /4 = 3.25 Floor: 13 //4 = 3

User Input : enables user to input their own data
-useful for surveys, gathering data for calculations
-can modify the data collected to represent integers ans floats instead of strings using the int and float functions.

Arithmetic operators are used to perform operations/calculations on integers and floats.
The addition and multiplication operators can also be used to combine 2 strings to repeat one for a number of times.

Comments are used to explain code
-single line : use #
-multi line : use triple quotes or # on every line

More Arithmetic operators:
Exponentiation(**): to the power of
Floor division: (//): standard division but rounds down to whole number
Modulus:(%):returns remainder

these can only be used on ints and floats

Thank You!

Any Questions?